Linux `ls` Command Notes

The `ls` command is used to list files and directories in Linux. It has multiple options to display details in different formats.

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1. Basic `ls`

```
ls
```

Description: Lists directory contents (files and folders).

Example:

```
ls
```

Output:

```
Documents  Downloads  file.txt  script.sh
```

---

2. Long Format `ls -l`

```
ls -l
```

Description: Displays files with details like permissions, owner, group, size, and modification date.

Example:

```
ls -l
```

Output:

```
-rw-r--r-- 1 prashant users  1024 Sep 15 21:00 file.txt
drwxr-xr-x 2 prashant users  4096 Sep 15 20:50 Documents
```

---

3. Show Hidden Files `ls -a`

```
ls -a
```

Description: Shows all files including hidden files (those starting with `.`).

Example:

```
ls -a
```

Output:

```
.  ..  .bashrc  .profile  Documents  file.txt
```

---

4. Long Format + Hidden `ls -la`

```
ls -la
```

Description: Combines `-l` and `-a` to show details of hidden and visible files.

Example:

```
ls -la
```

Output:

```
drwxr-xr-x  5 prashant users  4096 Sep 15 21:10 .
drwxr-xr-x 20 prashant users  4096 Sep 10 10:00 ..
-rw-r--r--  1 prashant users  1024 Sep 15 21:00 file.txt
```

---

5. Human-Readable Sizes `ls -lh`

```
ls -lh
```

Description: Displays file sizes in KB, MB, GB instead of raw bytes.

Example:

```
ls -lh
```

Output:

```
-rw-r--r-- 1 prashant users 1.0K Sep 15 21:00 file.txt
drwxr-xr-x 2 prashant users 4.0K Sep 15 20:50 Documents
```

---

6. Sort by Modification Time `ls -lt`

```
ls -lt
```

Description: Lists files sorted by time, newest first.

Example:

```
ls -lt
```

Output:

```
-rw-r--r-- 1 prashant users 1024 Sep 15 21:00 file.txt
drwxr-xr-x 2 prashant users 4096 Sep 15 20:50 Documents
```

---

7. Sort by Size `ls -lS`

```
ls -lS
```

Description: Lists files sorted by size (largest first).

Example:

```
ls -lS
```

Output:

```
-rw-r--r-- 1 prashant users 2048 Sep 15 21:00 bigfile.log
-rw-r--r-- 1 prashant users 1024 Sep 15 21:00 file.txt
drwxr-xr-x 2 prashant users 4096 Sep 15 20:50 Documents
```

Show File Sizes in Blocks

```
ls -s
```

Output:

```
4 file.txt
8 Documents
```

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8. Reverse Sorting `ls -lr` / `ls -ltr`

```
ls -lr   # Reverse alphabetical
ls -ltr  # Oldest files first
```

Example (ls -ltr):

```
drwxr-xr-x 2 prashant users 4096 Sep 10 10:00 OldFolder
-rw-r--r-- 1 prashant users 1024 Sep 15 21:00 file.txt
```

---

9. List Only Directories `ls -d */`

```
ls -d */
```

Description: Shows only directories.

Examples:

```
ls -d */
```

Output:

```
Documents/  Downloads/
```

```
ls -d /var
```

Output:

```
/var
```

```
ls -d .*   # hidden directories
```

Output:

```
.bashrc  .profile  .notes
```

---

10. Recursive Listing `ls -R`

```
ls -R
```

Description: Lists files recursively, including subdirectories.

Example:

```
ls -R
```

Output:

```
.:
Documents  Downloads  file.txt

./Documents:
doc1.txt  doc2.txt

./Downloads:
setup.exe
```

---

11. Combine Multiple Options

```
ls -alhtr
```

Meaning: - `-a` → Show hidden files - `-l` → Long format - `-h` → Human-readable size - `-t` → Sort by time - `-r` → Reverse order

Example:

```
ls -alhtr
```

Output:

```
-rw-r--r-- 1 prashant users 1.0K Sep 10 10:00 oldfile.txt
drwxr-xr-x 2 prashant users 4.0K Sep 15 20:50 Documents
-rw-r--r-- 1 prashant users 1.0K Sep 15 21:00 file.txt
```

---

12. Enable Colorized Output

```
ls --color
```

Description: Displays files with color-coded output.

Example:

```
ls --color
```

Output:

```
Documents  Downloads  file.txt  script.sh
```

(with colors depending on file type)

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13. Pattern Matching with `ls`

```
ls *.txt     # all files ending with .txt
ls file*     # all files starting with "file"
```

Output:

```
file.txt  file1.txt
```

Special Entries in `ls`

- `.` → Current directory
- `..` → Parent directory

Example:

```
ls -a
```

Output (if no hidden files exist):

```
.   ..   file1.txt   file2.txt   Projects/
```

Output (if hidden files exist):

```
.  ..  .bashrc  .profile  .notes  file1.txt  file2.txt  Projects/
```

Quick Summary of Common Commands

```
ls          # List files
ls -a       # Show hidden files
ls -l       # Long listing
ls -la      # Long + hidden
ls -lh      # Human-readable sizes
ls -lt      # Sort by time (newest first)
ls -lS      # Sort by size (largest first)
ls -lr      # Reverse sort
ls -d */    # List only directories
ls -R       # Recursive listing
ls -alhtr   # Combined options
ls --color  # Colored output
ls *.txt    # Match files by pattern
ls file*    # Match starting with 'file'
```

✅These are the **most useful** `ls` **command variations** with explanations and examples.